Name_____ Lab_____ Instructor_____

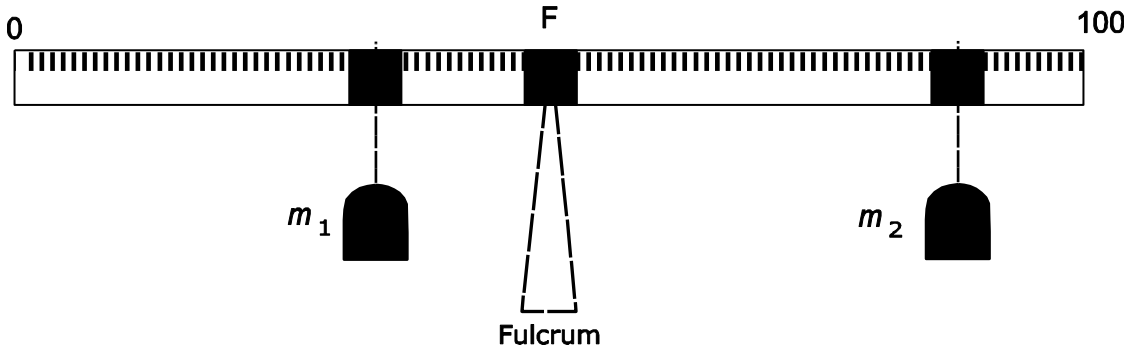
Pre-lab Assignment - Experiment 4

Equilibrium and Center of Gravity of a Rigid Body

1. (2 pts) Given the arrangement of masses hanging from a meter stick shown in the figure below (which is in static equilibrium), what can one say about the relative magnitude of the masses?

a) $m_1 > m_2$ b) $m_1 = m_2$ c) $m_1 < m_2$

Explain:



2. (2 pts) Suppose you have a meter stick balanced at its midpoint on a fulcrum as shown in the lab write up and the figure above. If mass m_1 = 25 g is placed 20 cm from the balance point and mass m_2 is 35 cm from the balance point when the meter stick is horizontal, what is the mass m_2?

3. (2 pts) What is the torque about the midpoint of the meter stick due to mass m_1?

Name _____ Lab _____ Instructor _____

Experiment 4

Equilibrium and Center of Gravity of a Rigid Body

Object:

To study the conditions for equilibrium of a horizontal meter stick under the action of gravitational forces and find the center of gravity of a non-uniform bar.

Apparatus:

Meter stick, fulcrum and support, hanger type masses, beam balance, unknown weight, loaded meter stick.

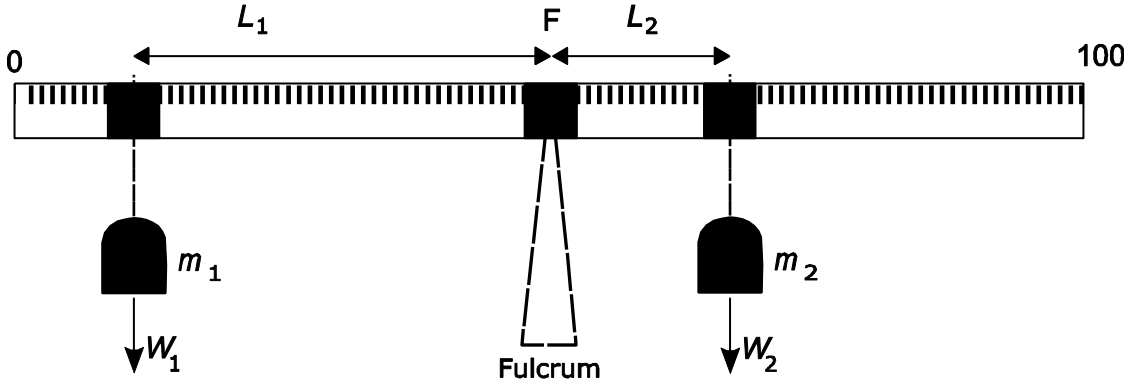


Figure 1- Meter stick with masses

Background:

The second condition of static equilibrium states that:

$$\sum \tau = 0$$

where clockwise (CW) torques are traditionally negative, thus:

$$\sum \tau_{CW} = \sum \tau_{CCW} \quad \text{or } L_2 W_2 = L_1 W_1$$

Procedure A:

We will be using masses calibrated in grams to hang on the meter stick. The force of gravity, or the weight, W, can be calculated in Newtons by multiplying the mass in kilograms by the accelerations due to gravity, $g = 9.80$ m/s^2.

Part 1: Clamp the knife edge at the point where the meter stick exactly balances. Make sure that the knife edge clip is placed on the stick so that the support is near the top of the bar and the cm marks on the bar are pointing up with the zero on the left side (see Figure 1). This means that the center of gravity of the meter stick will be below the point of suspension on the fulcrum. When the meter stick is balanced, one end will be just as far above the table top as the other end. Weigh the meter stick, recording the mass and the point of balance in Table 1. Place a single mass on one

side of the meter stick and balance it with a different mass on the opposite side. If you do not have hanger-type masses, use a loop of string with "negligible mass" to hang the large masses from the meter stick. Record the magnitude of each force in Newtons and its location on the bar in meters. Record the data under Part 1 in Table 1.

Part 2: Repeat the above procedure with the masses at different positions and record your data under Part 2.

Part 3: Now hang two masses at ***different locations*** on one side of the fulcrum and balance with one mass on the opposite side. Record the data under Part 3.

Record all data and calculations as indicated in Table 1. Some spaces in the torque columns will not be filled since a force cannot produce a clockwise and counterclockwise torque at the same time.

Use the pivot point (point of balance) as the axis of rotation, or center of torques, in your calculations.

Table 1

Mass of meter stick _____kg Point of balance of meter stick _____ m

** For percent difference in torques, take the percent difference between the CW torques and the CCW torques for each trial.

Part	Forces (N)	Position on bar (m)	Lever Arm (m)	Clockwise Torque (N·m)	Counterclockwise Torque (N·m)	Percent Difference in Torques **
1						
2						
3						

Part 4: Hang the unknown mass from the left side (L_1) and balance it with a known mass. Obtain the mass of the unknown with the beam balance. Record your data here

L_1 = _____

For known mass: m_2 = _____ W_2 = _____ L_2 = _____

Remember counterclockwise torques are positive and clockwise torques are negative. Using the $\Sigma\tau = 0$, equation, write out all torques acting on the meter stick treating W_1 as an unknown. Solve for W_1. Record the mass $M_1 = W_1/g$ and compare it to the value obtained using the scale.

Show your calculation for M_1 using the data from Part 4:

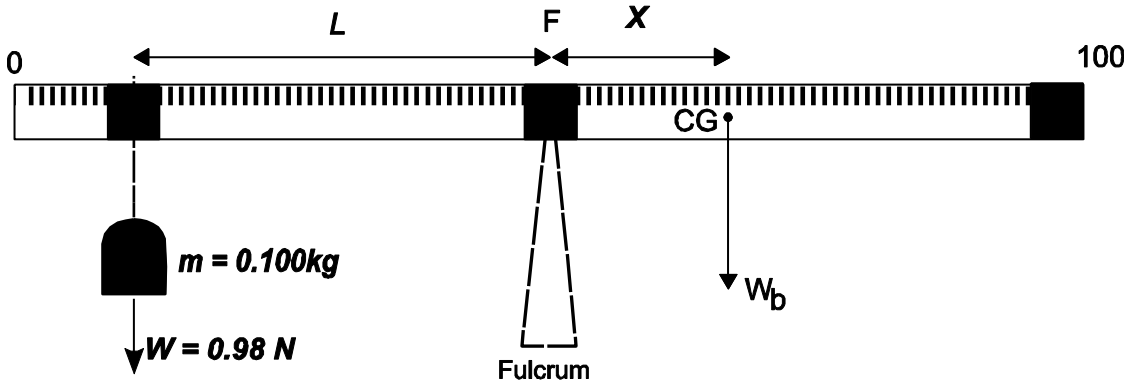
M_1 (calculated) _____ M_1 (using scale) _____

% difference _____

Procedure B:

A meter bar loaded at one end is provided for use in this part of the experiment. It is the purpose of this part of the experiment to locate the center of gravity of the loaded meter bar by different methods. First obtain the mass of the bar using the digital scale. Convert this mass to kg. Multiply by 9.80 m/s^2 to find the weight of the loaded meter stick. Record this as W_b in Table 2.

Balance the loaded meter stick on the fulcrum. Record this balance position as the center of gravity CG in Table 2.



Now shift the fulcrum, F, to some convenient position as shown in Figure 2 and balance the bar with the 0.100 kg mass. Only one mass (the 0.100 kg) will be hanging from the bar when it is balanced. The weight of the 0.100 kg mass at a distance, L, from the fulcrum now balances the weight of the bar, W_b, considered to be located at the center of gravity, CG, at a distance X from the fulcrum. Record the fulcrum position, L, W_b, and your measured value of X in Data Table 2. Using $\Sigma\tau = 0$ gives:

$$(0.980\ \text{N})L - W_b X = 0$$

Solve for X using your data and the above equation, record its value in Data Table 2. Compare it to your measured value of X using a percentage difference calculation with the measured value of X as your standard value.

Repeat the procedure for a different position of the fulcrum.

Table 2

Weight of loaded meter stick, W_b = _____ N

Point of Balance of stick alone, CG = _____ m

Trial	Position of Fulcrum (m)	Distance of 0.10-kg mass from Fulcrum L(m)	Calculated X (m)	Measured X (m)	% Difference in X
1					
2					

Questions:

(1) In part A-3, there are 4 forces acting on the stick: gravity(directly) and the forces from the three hanging masses. For each of the 4 forces, calculate the torque from each force using the ***zero end*** of the stick as the axis of rotation. Show your work for each torque. Do the four torques roughly add up to zero still? (Be careful with the signs of the torques).

(2) How could a person riding in the weightless environment of the International Space Station (i.e. in microgravity, not on the moon) find the center of mass of the non-uniform bar?